

Subject-Level Calibration Heterogeneity is Stable Across LLM Families and Scales: A Statistical Audit of MMLU

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Introduction

- LLM confidence is of significant interest in addition to accuracy.
- Calibration is usually reported as a scalar (typically Expected Calibration Error, ECE), possibly obscuring underlying structure:

$$\text{ECE} = \sum_b \frac{|B_b|}{n} |\text{acc}(B_b) - \text{conf}(B_b)|$$

Questions:

- How does calibration vary across QA domains?
- Is this variation consistent across models?
- How is calibration associated with other features (scale, question-level covariates)?

Data & Methods

Data. Questions: MMLU: 14,042 four-option questions across 57 subjects nested in 4 domains (STEM, Social Sciences, Humanities, Other). Subject sizes 100–1,534 (median 173).
Confidence elicitation: renormalized softmax over the four answer tokens:

$$c_i = \max_{a \in \{A, B, C, D\}} \frac{e^{\ell_a}}{\sum_{a'} e^{\ell_{a'}}}$$

Models: GPT-4o-mini, Llama-3-8B-Instruct, and Qwen-2.5-{0.5B, 1.5B, 3B, 7B, 14B}-Instruct. The outcome is the signed *calibration gap* $\text{gap}_i = c_i - y_i$, with $y_i \in \{0, 1\}$ indicating correctness (positive under overconfidence).
Three-level mixed model (questions $\rightarrow$ subjects $\rightarrow$ domains):

$$\text{gap}_{ijk} = \mu + \mathbf{x}_{ijk}^\top \boldsymbol{\beta} + u_k + v_{jk} + \varepsilon_{ijk},$$

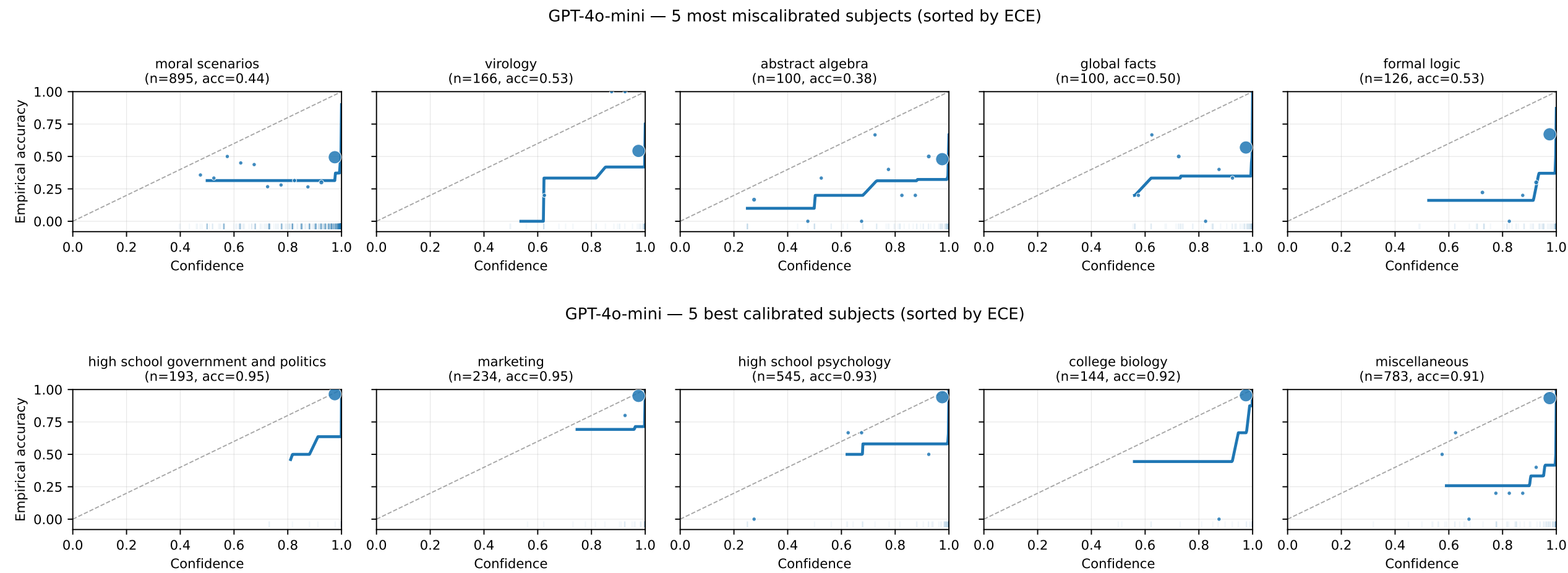
fit using REML.

Also: per-subject one-sample t -tests with Benjamini–Hochberg FDR control at $\alpha = 0.05$; cross-model Spearman correlations of subject-level ECE with bootstrap CIs.

Results

Every model is overconfident. The mean calibration gap (mixed-model intercept) is positive for all seven models, ranging 0.18–0.31. This is consistent with prior work (Guo et al. 2017; Kadavath et al. 2022).

Calibration across subjects Across the six larger models, virology and professional law appear in the top-10 most miscalibrated for every model; a cluster of college-level technical subjects (abstract algebra, college mathematics, machine learning) is in the top-5 for four of six. Symmetrically, marketing and high school psychology are best-calibrated for every model. Specialization level tracks miscalibration: high-school subjects show systematically smaller gaps than their college-level counterparts.



GPT-4o-mini reliability curves: the five most miscalibrated (top) and five best-calibrated (bottom) subjects. Solid: isotonic fit on observed support; markers: 20-bin empirical accuracy sized by count; dashed: perfect calibration.

Variance decomposition

Most calibration-gap variance is at the question level (92.0–98.5%) and domain variance collapses to zero; between-subject ICC spans nearly $5\times$ across models.

Model	$\sigma_{\text{subject}}^2$	ICC _{subject}
GPT-4o-mini	0.013	0.079
Llama-3-8B	0.004	0.020
Qwen-2.5-0.5B	0.004	0.018
Qwen-2.5-1.5B	0.003	0.017
Qwen-2.5-3B	0.011	0.051
Qwen-2.5-7B	0.009	0.049
Qwen-2.5-14B	0.010	0.059

ICC_{subject}: share of total calibration-gap variance at the subject level. Domain variance is estimated at zero for all models.

FDR-controlled miscalibration testing

For each of the 57 subjects s we test whether its mean calibration gap is zero,

$$H_0 : \mathbb{E}[\text{gap}_i \mid \text{subject } s] = 0 \quad \text{vs.} \quad H_1 : \mathbb{E}[\text{gap}_i \mid \text{subject } s] \neq 0,$$

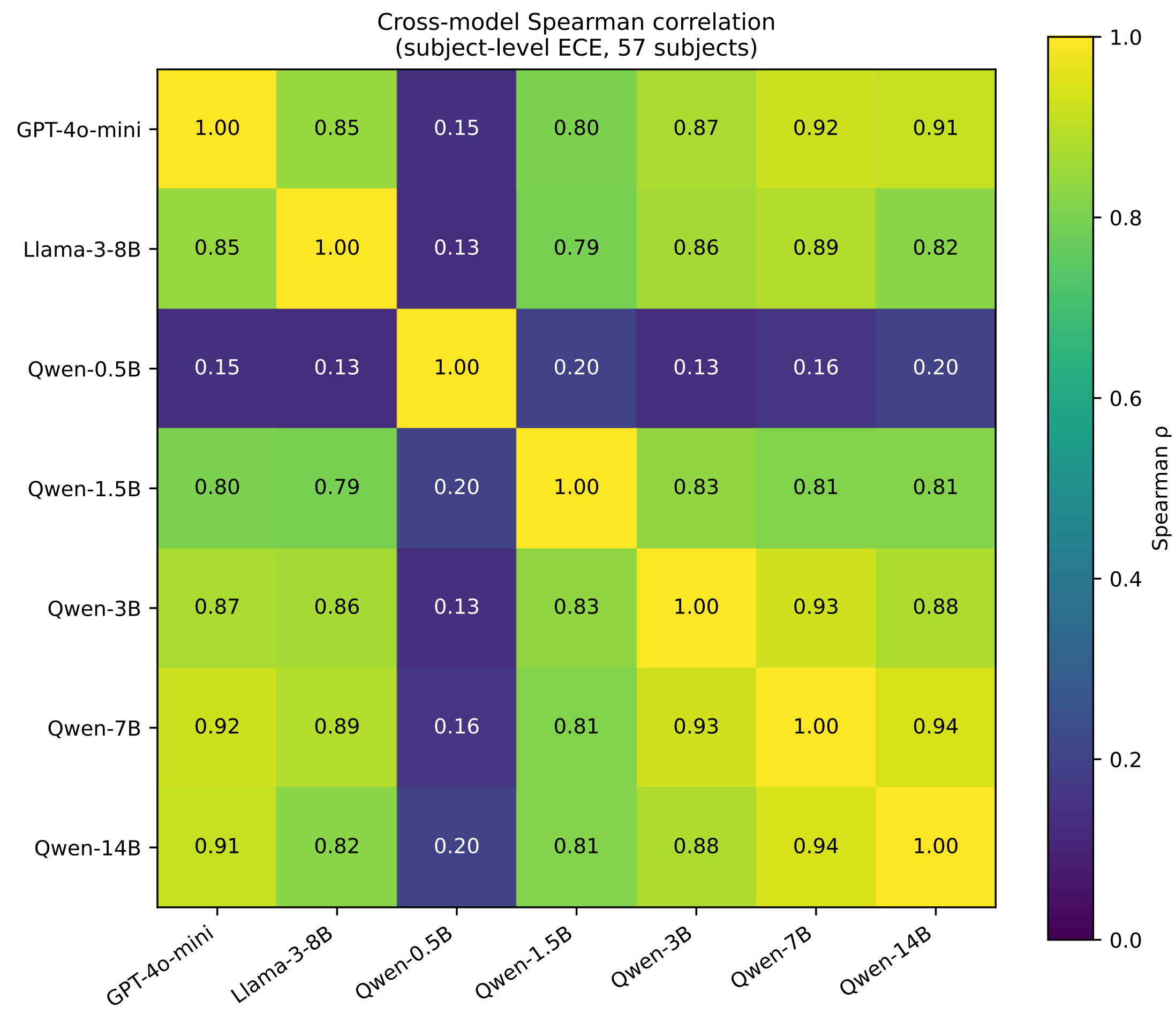
with a two-sided one-sample t -test, controlling the false discovery rate (FDR) at $\alpha = 0.05$ via Benjamini–Hochberg (1995). Rejections are near-universal (the five largest models reject all 57 subjects), so the story is the *magnitude* of the gaps, not the binary outcome: best- to worst-calibrated gap ranges span $3.8\text{--}13.0\times$ across the six larger models.

Robustness Checks

ECE rankings are stable across bin counts (Spearman ≥ 0.97 vs. the 20-bin baseline) and agree with a strictly proper scoring rule, the Negative Log Calibration Score (Spearman ≥ 0.79); two- and three-level ICC estimates are numerically identical.

Cross-model & cross-family stability

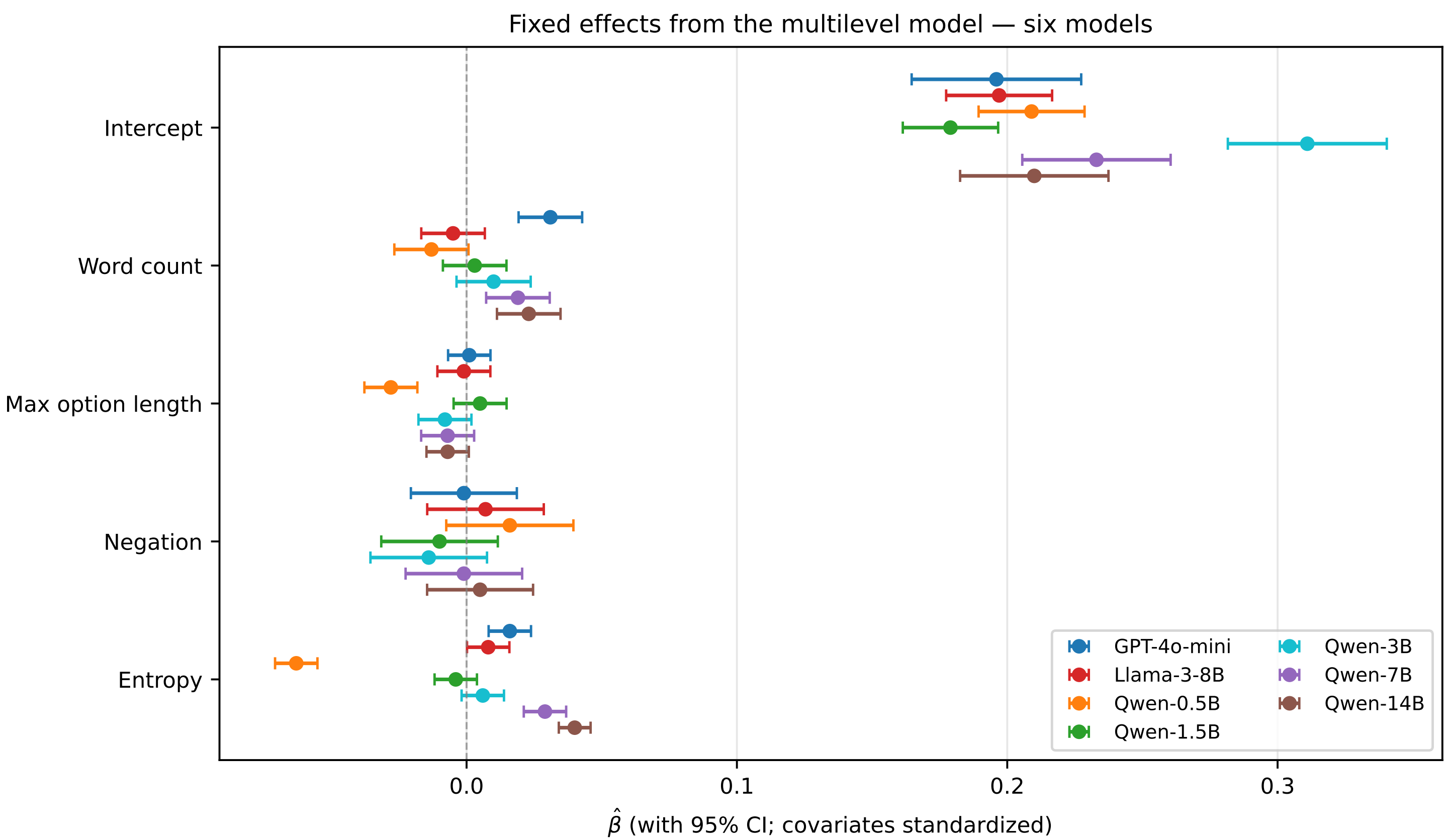
All 15 pairwise Spearman correlations among the six larger models fall in $[0.80, 0.94]$. Cross-family GPT-4o-mini vs. Qwen-7B/14B (0.92/0.91) is as strong as the best within-family pair (Qwen-7B vs. 14B, 0.94). Qwen-0.5B is a conspicuous exception ($\rho \in [0.13, 0.20]$), driven by $10.6\times$ positional bias toward D—a floor below which our elicitation protocol breaks down.



Spearman rank correlation of subject-level ECE across the seven models (57 common subjects).

Question-level predictors

The three-level mixed model relates the calibration gap to standardized question features. Word count positively predicts overconfidence for GPT-4o-mini and the larger Qwens; negation and option length are mostly inert. The most informative pattern is the **entropy** coefficient (next panel).



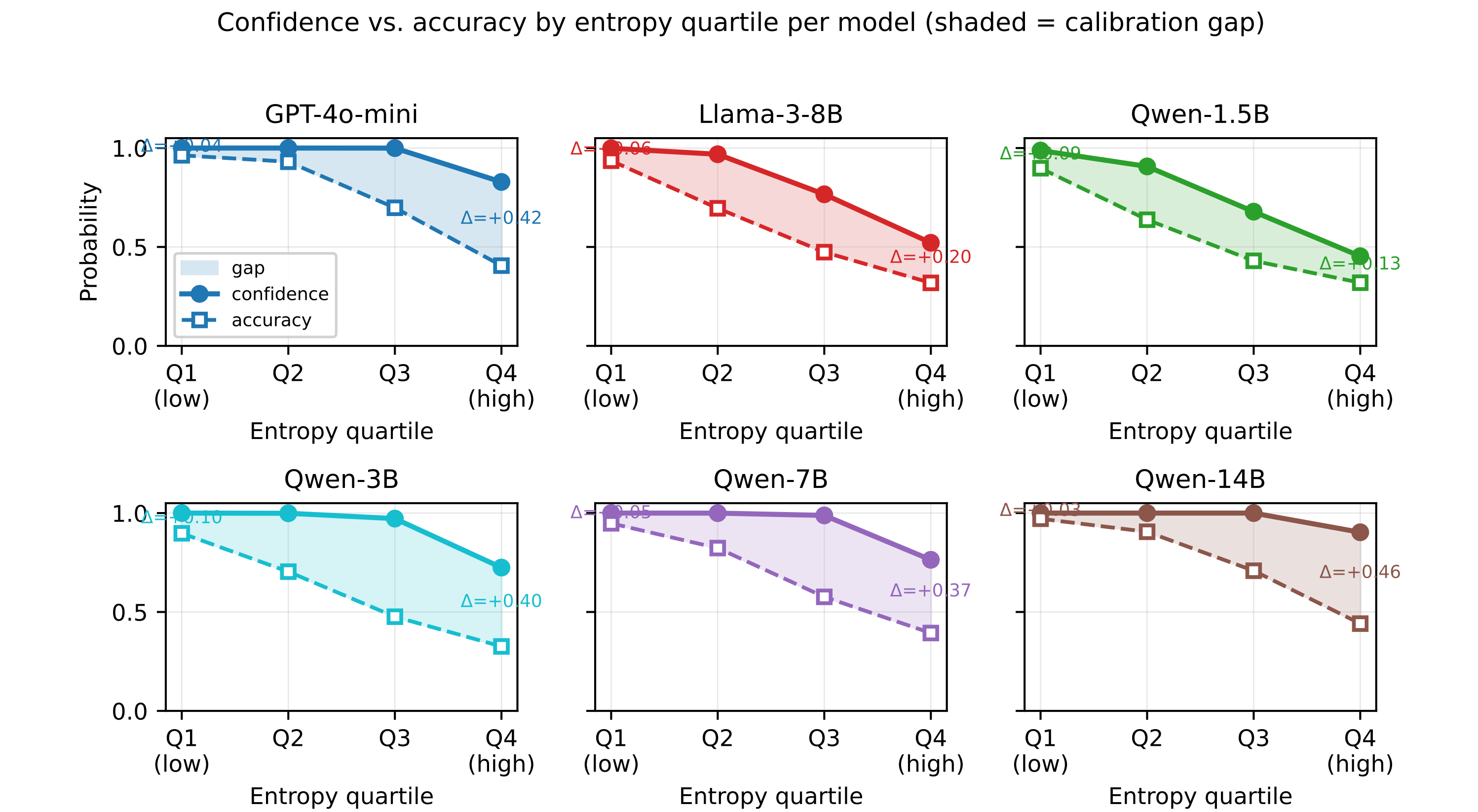
Fixed-effect estimates ($\pm 95\%$ CI) on the standardized calibration-gap scale. The entropy coefficient shows a clear sign progression across the Qwen scaling sweep.

Difficulty, entropy, and scale

Entropy H_i of the A/B/C/D distribution proxies how hard the model finds an item. Naively, higher entropy should shrink the gap (top-1 confidence is mechanically lower). Instead, within the matched-precision Qwen sweep (1.5B fp16, 3B, 7B, 14B) the entropy coefficient grows monotonically:

$$-0.004 \rightarrow +0.006 \rightarrow +0.029 \rightarrow +0.040.$$

At 7B/14B, confidence stays near saturation across the first three entropy quartiles while accuracy keeps dropping, so the gap widens on the model's hardest items (Q1 vs. Q4 gap: $+0.05$ vs. $+0.37$ at 7B; $+0.03$ vs. $+0.46$ at 14B). The between-subject ICC also jumps from 1.5B to 3B ($0.017 \rightarrow 0.051$) and is flat thereafter— consistent with RLHF-induced confidence saturation (Kadavath et al. 2022; Nakkiran et al. 2025).



Confidence (solid) and accuracy (dashed) by entropy quartile; shaded area is the calibration gap.

Conclusion

Subject-level miscalibration is primarily a property of the task, not of any single training pipeline: it is large and stable across LLM families above $\sim 1\text{B}$ parameters (all 15 larger-model pairs $\rho \in [0.80, 0.94]$). Applied/factual high-school subjects are reliably well-calibrated; college-level quantitative and niche professional content are reliably bad. This cross-model stability is the precondition that makes subject-aware recalibration potentially worthwhile a global fix would under-correct the worst cluster (professional law, virology, college-level technical content) while over-correcting subjects like marketing.

References

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